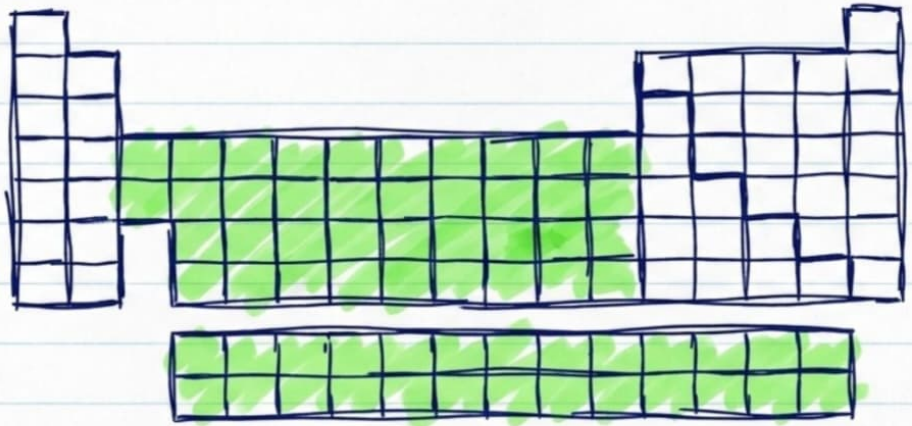
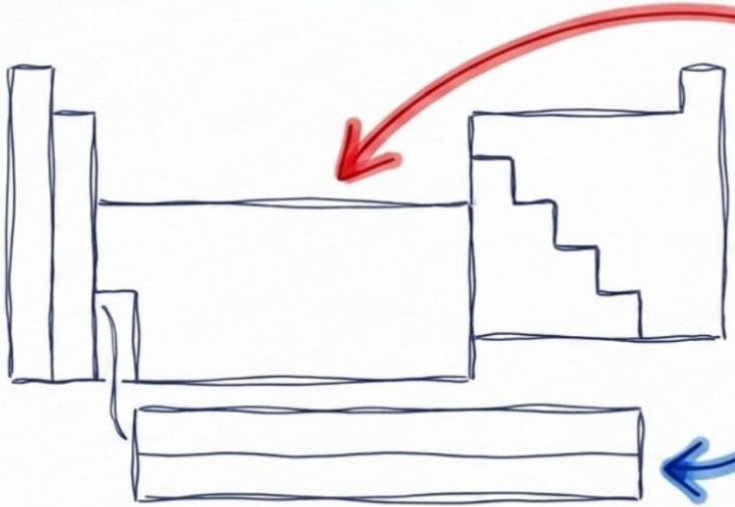
Subject: Chem Revision
Topic: Transition Metals

Ultimate Revision Notes: The d-block & f-block Elements

(Transition & Inner Transition Metals)





d-block (Transition Metals): Groups 3-12. We are progressively filling the d-orbitals across four long periods. These are the classic metals we build our world with (Fe, Cu, Ag, Au).

f-block (Inner Transition Metals): The bottom 'island'. We are filling the 4f and 5f orbitals. Pulled out just to keep the periodic table from being too too wide!

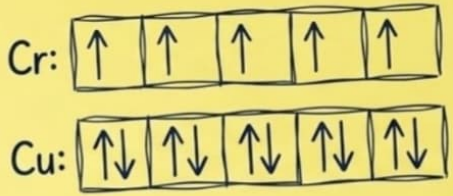
* Warning! Zn, Cd, and Hg are technically in the d-block, but they have completely full d-orbitals (d^{10}) in their ground and common oxidation states. Therefore, they are not regarded as true transition elements!

The General Rule: $(n-1)d^{1-10}\ ns^{1-2}$

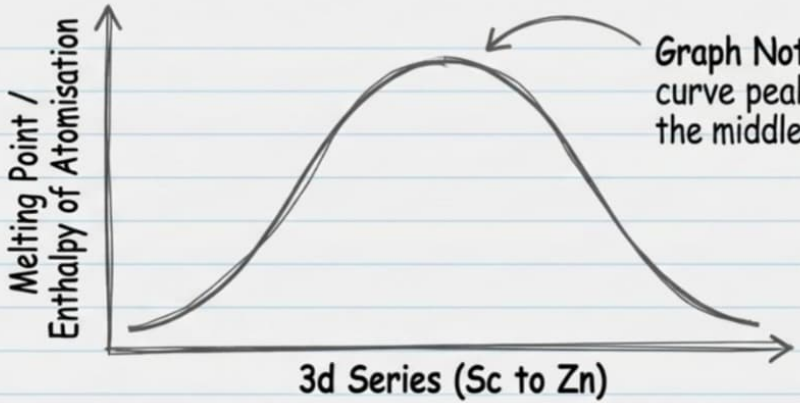
WARNING: Special Cases!

Chromium (Cr): Expect $3d^4\ 4s^2 \rightarrow$ Reality: $3d^5\ 4s^1$

Copper (Cu): Expect $3d^9\ 4s^2 \rightarrow$ Reality: $3d^{10}\ 4s^1$



Why? Half-filled (d^5) and fully-filled (d^{10}) sets of orbitals are relatively more stable. The energy gap between 4s and 3d is so tiny that an electron easily jumps over to reach perfect stability!



Graph Note: The curve peaks right in the middle (around d^5).

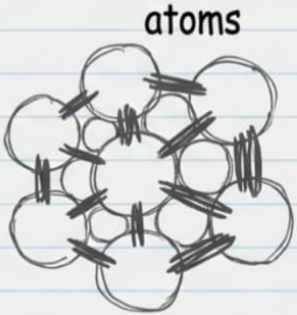
Why are they so Metallic?

Very hard, high Melting/Boiling Points, high Enthalpies of Atomisation.

The Secret: Unpaired electrons!

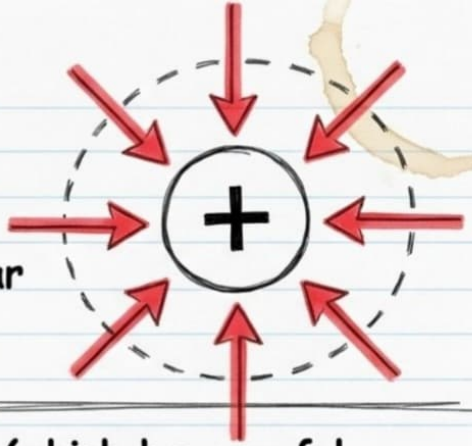
In transition metals, both the ns AND the inner (n-1)d electrons get involved in interatomic metallic bonding.

More unpaired d-electrons = stronger interatomic interaction = stronger bonds. That's why properties peak in the middle of the row!



1. <u>The Size Trend</u>: As you move across a row, atomic size decreases.

Why? Poor Shielding – New electrons go into inner d or f orbitals. These are terrible at shielding outer electrons from the growing nuclear charge. The nucleus pulls everything in tighter!



2. The Lanthanoid Contraction: Filling 4f orbitals (which have awful shielding) before the 5d series causes a massive size squeeze.

The Result: The 2nd and 3rd row transition metals are almost exactly the same size.

Zr (160 pm) Hf (159 pm)

Chemical Twins! Almost impossible to separate.

Superpower 1: Multiple Oxidation States

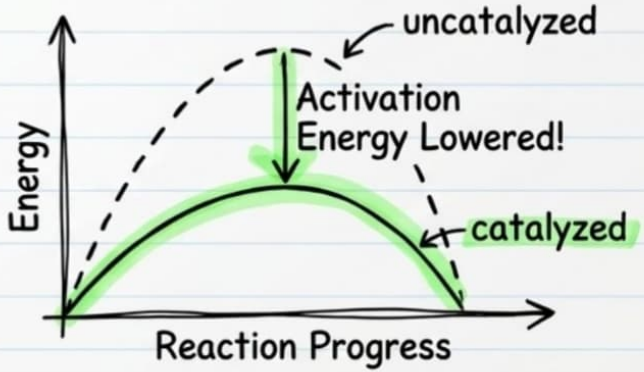
They lose varying numbers of electrons because the energies of (n-1)d and ns orbitals are very close.

The Crown Jewel: Manganese (Mn). It exhibits states all the way from +2 to +7!

Superpower 2: Elite Catalysts

They speed up reactions. Examples: Finely divided Iron (Fe) in Haber Process, V_2O_5 in Contact Process.

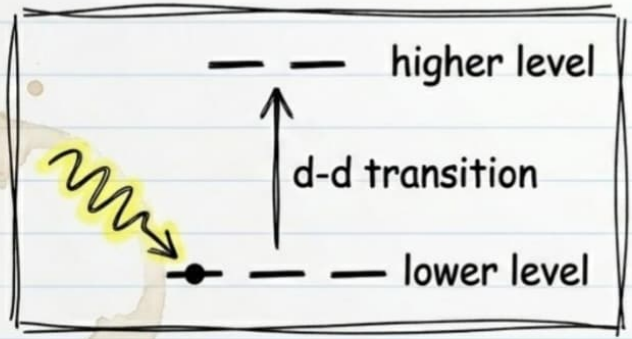
Why? They can easily change oxidation states and use empty d-orbitals to form temporary bonds with reactants, lowering activation energy.



Superpower 3: Coloured Ions

When ligands attach, the 5 d-orbitals split into two different energy levels.

An electron absorbs a photon of visible light and "jumps" (d-d transition) to the higher level. The colour we see is the light not absorbed!



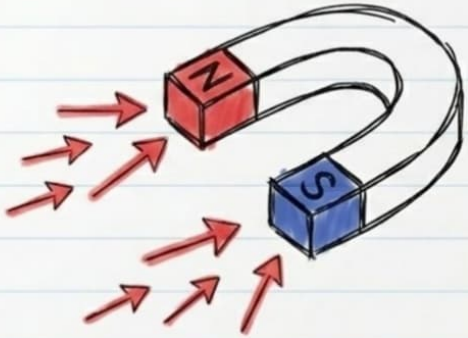
Superpower 4: Paramagnetism

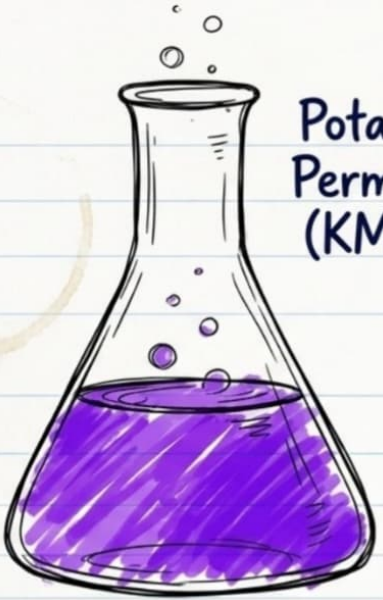
Attracted by a magnetic field.
Entirely caused by unpaired electrons.

The Spin-Only Formula:

$$\mu = \sqrt{n(n+2)}\ \text{BM}$$

(where n = number of unpaired electrons)





Potassium Permanganate ($KMnO_4$)

- The Purple Powerhouse! Very strong oxidising agent (used in textiles/bleach).
- Prep: Fusing Pyrolusite ore (MnO_2) with an alkali metal hydroxide and an oxidising agent like KNO_3.

Potassium Dichromate ($K_2Cr_2O_7$)

- The Orange Oxidiser! Strong oxidising agent (favourite for organic chemistry).
- Prep: Roast Chromite ore with sodium carbonate in air, extract, then acidify the solution.

Lanthanoids (4f)

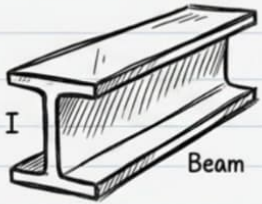
- Silvery-white, soft metals.
- +3 is the absolute favourite and most stable oxidation state.
- Chemistry is very similar across the entire row (thanks to the Lanthanoid contraction!).

Actinoids (5f)

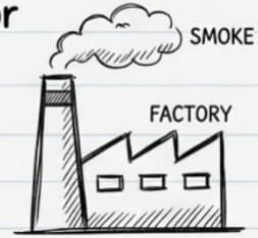
- Radioactive and tricky to study!
- Way more complex: They show a huge range of oxidation states (from +3 all the way up to +7 in early elements).
- Why the difference? 5f electrons aren't buried as deep inside the atom as 4f electrons, so they participate much more in bonding.

Transition Metals in the Real World

Alloys & Steels: Because their atoms are similar sizes, they mix perfectly! Cr, V, W, and Mn are mixed with Iron to make hard, high-melting stainless steels.



Industrial Catalysts: Iron (ammonia production), Nickel (hydrogenation of oils), Titanium (TiO_2 for white pigments).



Everyday Power: Zinc & Nickel/Cadmium are the lifeblood of battery cells.

Coinage: Group 11 (Copper, Silver, Gold) – the noble metals!

Good luck on the exam! :)